Department of Mathematics and Natural Sciences

Final Examination
Semester: Spring 2025
Course Title: MATHEMATICS I: Differential Calculus & Coordinate Geometry
Course Code: MAT-110
Time: 1 hour 20 minutes **Date: May 19, 2025**
Total Marks: 35

SOLUTION & MARKING SCHEME

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✅ NOTE:

TO APPRECIATE YOUR MATHEMATICAL THINKING, ANY VALID MATHEMATICAL WAY OF SOLVING WILL BE TREATED EQUALLY AND ASSESSED ACCORDINGLY, AS LONG AS IT'S CONSISTENT WITH THE PROVIDED INSTRUCTIONS OF THE QUESTION.

IN THIS ASSESSMENT PROCESS EVERY PROBLEM WILL BE ASSESSED BASED ON 3 CRITERIA:
A. MATHEMATICAL THINKING
B. MATHEMATICAL KNOWLEDGE
C. MATHEMATICAL SKILL

✅ PERCENTAGE SCHEME

- **N/A = Not Assessed / Not Answered**
- **0% = Incorrect**
- **50% = Partially Correct**
- **80% = Mostly Correct**
- **100% = Perfect**

Q1. Use appropriate forms of the chain rule to find $\frac{\partial w}{\partial u}$ and $\frac{\partial w}{\partial v}$ for [7]

$$w = rs^2 \ln t,\ r = u^2 \sin v,\ s = 4v + uv \text{ and } t = 3u^2 v.$$

Solution:

Step 1: Use the **chain rule.**

Since $w = rs^2 \ln t$, and all three variables r, s, t depend on u and v, apply the multivariable chain rule:

$$\frac{\partial w}{\partial u} = \frac{\partial w}{\partial r}\frac{\partial r}{\partial u} + \frac{\partial w}{\partial s}\frac{\partial s}{\partial u} + \frac{\partial w}{\partial t}\frac{\partial t}{\partial u}$$

Similarly,

$$\frac{\partial w}{\partial v} = \frac{\partial w}{\partial r}\frac{\partial r}{\partial v} + \frac{\partial w}{\partial s}\frac{\partial s}{\partial v} + \frac{\partial w}{\partial t}\frac{\partial t}{\partial v}$$

Step 1: Partial derivatives of w

$$\frac{\partial w}{\partial r} = s^2 \ln t, \quad \frac{\partial w}{\partial s} = 2rs \ln t, \quad \frac{\partial w}{\partial t} = \frac{rs^2}{t}$$

Step 2: Partials of intermediate variables

$$\begin{aligned} \frac{\partial r}{\partial u} &= 2u \sin v, & \frac{\partial r}{\partial v} &= u^2 \cos v \\ \frac{\partial s}{\partial u} &= v, & \frac{\partial s}{\partial v} &= 4 + u \\ \frac{\partial t}{\partial u} &= 6uv, & \frac{\partial t}{\partial v} &= 3u^2 \end{aligned}$$

Step 3: Apply the multivariable chain rule

$$\boxed{\frac{\partial w}{\partial u} = (s^2 \ln t)(2u \sin v) + (2rs \ln t)(v) + \left(\frac{rs^2}{t}\right)(6uv)}$$

$$\boxed{\frac{\partial w}{\partial v} = (s^2 \ln t)(u^2 \cos v) + (2rs \ln t)(4 + u) + \left(\frac{rs^2}{t}\right)(3u^2)}$$

Q2. Determine the 1st degree ($L(x,y)$) and 2nd degree ($Q(x,y)$) Taylor polynomial approximations respectively of the function, $f(x,y) = \ln(x^2 + y^2 + 1)$, near the point $(1,1)$. [7]

Solution:

Given:

$$f(x,y) = \ln(x^2 + y^2 + 1), \quad (a,b) = (1,1)$$

Solution:

$$f(1,1) = \ln(1^2 + 1^2 + 1) = \ln(3)$$

First partial derivatives:

$$f_x = \frac{2x}{x^2+y^2+1}, \quad f_y = \frac{2y}{x^2+y^2+1}$$
$$f_x(1,1) = \frac{2}{3}, \quad f_y(1,1) = \frac{2}{3}$$

Linear approximation:

$$L(x,y) = \ln(3) + \frac{2}{3}(x-1) + \frac{2}{3}(y-1)$$

Second partial derivatives:

$$f_{xx} = \frac{2(x^2+y^2+1)-4x^2}{(x^2+y^2+1)^2}, f_{yy} = \frac{2(x^2+y^2+1)-4y^2}{(x^2+y^2+1)^2}, f_{xy} = -\frac{4xy}{(x^2+y^2+1)^2}$$

At (1,1):

$$f_{xx} = \frac{2(3)-4(1)}{9} = \frac{6-4}{9} = \frac{2}{9}, f_{yy} = \frac{2(3)-4(1)}{9} = \frac{6-4}{9} = \frac{2}{9}, f_{xy} = -\frac{4}{9}$$

Second-degree (Quadratic):

$$Q(x,y) = L(x,y) + \frac{1}{2}\left[\frac{2}{9}(x-1)^2 - \frac{8}{9}(x-1)(y-1) + \frac{2}{9}(y-1)^2\right]$$

Q3. Find all relative extrema and saddle points (if any) for the function: [7]

$$f(x,y) = 4x^2e^y - 2x^4 - e^{4y}.$$

Solution:

See here:

https://www.emathhelp.net/calculators/calculus-3/critical-points-extrema-saddle-points-calculator/?f=+4x%5E2+*+e%5Ey+-+2x%5E4+-+e%5E%284y%29

First, compute the first-order partial derivatives:

$$f_x = 8xe^y - 8x^3, \quad f_y = 4x^2e^y - 4e^{4y}$$

Set $f_x = 0$ and $f_y = 0$:

$$8x(e^y - x^2) = 0 \Rightarrow x = 0 \text{ or } e^y = x^2$$

$$x^2e^y = e^{4y} \Rightarrow x^2 = e^{3y}$$

From above:

$$e^y = x^2 \text{ and } x^2 = e^{3y} \Rightarrow e^y = e^{3y} \Rightarrow y = 0 \Rightarrow x^2 = 1 \Rightarrow x = \pm 1$$

So, critical points: $(1,0)$, $(-1,0)$

Second-order partial derivatives:

$$f_{xx} = 8e^y - 24x^2, \quad f_{yy} = 4x^2e^y - 16e^{4y}, \quad f_{xy} = 8xe^y$$

At $(1,0)$:

$$f_{xx} = -16, \quad f_{yy} = -12, \quad f_{xy} = 8$$

$$D = f_{xx}f_{yy} - (f_{xy})^2 = (-16)(-12) - 64 = 128$$

Since $D > 0$ and $f_{xx} < 0$, $(1,0)$ is a **local maximum.**

By symmetry, $(-1,0)$ is also a **local maximum.**

Q4. Use Lagrange multiplier to find maximum and minimum values of the function, [7]
$f(x,y,z) = 2x + y - 2z$ subject to $x^2 + y^2 + z^2 = 9$.

Solution:

See here:

https://www.emathhelp.net/calculators/calculus-3/lagrange-multipliers-calculator/?f=2x%2By-2z&g=x%5E2+%2By%5E2+%2Bz%5E2+%3D9&h=

SOLUTION

Attention! This calculator doesn't check the conditions for applying the method of Lagrange multipliers. Use it at your own risk: the answer may be incorrect.

Rewrite the constraint $x^2 + y^2 + z^2 = 9$ as $x^2 + y^2 + z^2 - 9 = 0$.

Form the Lagrangian: $L(x, y, z, \lambda) = (2x + y - 2z) + \lambda\left(x^2 + y^2 + z^2 - 9\right)$.

Find all the first-order partial derivatives:

$\frac{\partial}{\partial x}\left((2x + y - 2z) + \lambda\left(x^2 + y^2 + z^2 - 9\right)\right) = 2\lambda x + 2$ (for steps, see partial derivative calculator).

$\frac{\partial}{\partial y}\left((2x + y - 2z) + \lambda\left(x^2 + y^2 + z^2 - 9\right)\right) = 2\lambda y + 1$ (for steps, see partial derivative calculator).

$\frac{\partial}{\partial z}\left((2x + y - 2z) + \lambda\left(x^2 + y^2 + z^2 - 9\right)\right) = 2\lambda z - 2$ (for steps, see partial derivative calculator).

$\frac{\partial}{\partial \lambda}\left((2x + y - 2z) + \lambda\left(x^2 + y^2 + z^2 - 9\right)\right) = x^2 + y^2 + z^2 - 9$ (for steps, see partial derivative calculator).

Next, solve the system $\begin{cases} \frac{\partial L}{\partial x} = 0 \\ \frac{\partial L}{\partial y} = 0 \\ \frac{\partial L}{\partial z} = 0 \\ \frac{\partial L}{\partial \lambda} = 0 \end{cases}$, or $\begin{cases} 2\lambda x + 2 = 0 \\ 2\lambda y + 1 = 0 \\ 2\lambda z - 2 = 0 \\ x^2 + y^2 + z^2 - 9 = 0 \end{cases}$.

The system has the following real solutions: $(x, y, z) = (-2, -1, 2)$, $(x, y, z) = (2, 1, -2)$.

$f(-2, -1, 2) = -9$

$f(2, 1, -2) = 9$

Thus, the minimum value is -9, and the maximum value is 9.

Q5. Transform the conic $9y^2 - 8x + 4x^2 - 36y + 4 = 0$ into its standard form. Then find its eccentricity, foci, vertices, covertices, equations of directrices and length of latus rectum. [7]

Solution:

See here:
https://www.emathhelp.net/calculators/algebra-2/ellipse-calculator/?type=f&f=9y%5E2+-8x+%2B4x%5E2+-36y+%2B4&cx=&cy=&f1x=&f1y=&f2x=&f2y=&v1x=&v1y=&v2x=&v2y=&cv1x=&cv1y=&cv2x=&cv2y=&ma=&mi=&e=&a=&d1=&d2=&p1x=&p1y=&p2x=&p2y=&p3x=&p3y=&p4x=&p4y=

The equation of an ellipse is $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$, where (h, k) is the center, a and b are the lengths of the semi-major and the semi-minor axes.

Our ellipse in this form is $\frac{(x-1)^2}{9} + \frac{(y-2)^2}{4} = 1$.

Thus, $h = 1$, $k = 2$, $a = 3$, $b = 2$.

The standard form is $\frac{(x-1)^2}{3^2} + \frac{(y-2)^2}{2^2} = 1$.

The vertex form is $\frac{(x-1)^2}{9} + \frac{(y-2)^2}{4} = 1$.

The general form is $4x^2 - 8x + 9y^2 - 36y + 4 = 0$.

The eccentricity is $e = \frac{c}{a} = \frac{\sqrt{5}}{3}$.

The first focus is $(h - c, k) = \left(1 - \sqrt{5}, 2\right)$.

The second focus is $(h + c, k) = \left(1 + \sqrt{5}, 2\right)$.

The first vertex is $(h - a, k) = (-2, 2)$.

The second vertex is $(h + a, k) = (4, 2)$.

The first co-vertex is $(h, k - b) = (1, 0)$.

The second co-vertex is $(h, k + b) = (1, 4)$.

The first directrix is $x = h - \frac{a^2}{c} = 1 - \frac{9\sqrt{5}}{5}$.

The second directrix is $x = h + \frac{a^2}{c} = \frac{5+9\sqrt{5}}{5}$.

The length of the latera recta (focal width) is $\frac{2b^2}{a} = \frac{8}{3}$.

Q6. Compute f_{xy} and f_{yxz} for the function $f(x, y, z) = \sqrt{xy} + \ln(y^3 z^2) - x \tan z$. [7]
Also convert $(\rho, \theta, \varphi) = \left(5, \frac{4\pi}{3}, \frac{7\pi}{6}\right)$ into cylindrical coordinates and $(x, y, z) = (4, -4, 4\sqrt{6})$ into spherical coordinates.

Solution:
See here:
https://www.emathhelp.net/calculators/calculus-3/partial-derivative-calculator/?f=%28xy%29%5E0.5+%2B+ln%28y%5E3+z%5E2%29+-x+tan+z&var=x%2Cy

Your input: find $\frac{\partial^2}{\partial x \partial y}\left(-x \tan(z) + \sqrt{xy} + \ln\left(y^3 z^2\right)\right)$

Rewrite the logarithm:

$$\frac{\partial^2}{\partial x \partial y}\left(-x \tan(z) + \sqrt{xy} + \ln\left(y^3 z^2\right)\right) = \frac{\partial^2}{\partial x \partial y}\left(-x \tan(z) + \sqrt{xy} + 3\ln(y) + 2\ln(z)\right)$$

FIRST, FIND $\frac{\partial}{\partial x}\left(-x \tan(z) + \sqrt{xy} + 3\ln(y) + 2\ln(z)\right)$

The derivative of a sum/difference is the sum/difference of derivatives:

$$\frac{\partial}{\partial x}\left(-x \tan(z) + \sqrt{xy} + 3\ln(y) + 2\ln(z)\right) = \left(\frac{\partial}{\partial x}\left(\sqrt{xy}\right) - \frac{\partial}{\partial x}\left(x \tan(z)\right) + \frac{\partial}{\partial x}\left(3\ln(y)\right) + \frac{\partial}{\partial x}\left(2\ln(z)\right)\right)$$

The derivative of a constant is 0:

$$\frac{\partial}{\partial x}(3\ln(y)) + \frac{\partial}{\partial x}\left(\sqrt{xy}\right) - \frac{\partial}{\partial x}(x\tan(z)) + \frac{\partial}{\partial x}(2\ln(z)) = (0) + \frac{\partial}{\partial x}\left(\sqrt{xy}\right)$$
$$- \frac{\partial}{\partial x}(x\tan(z)) + \frac{\partial}{\partial x}(2\ln(z))$$

Apply the constant multiple rule $\frac{\partial}{\partial x}(c\cdot f) = c\cdot\frac{\partial}{\partial x}(f)$ with $c = \tan(z)$ and $f = x$:

$$-\frac{\partial}{\partial x}(x\tan(z)) + \frac{\partial}{\partial x}\left(\sqrt{xy}\right) + \frac{\partial}{\partial x}(2\ln(z)) = -\tan(z)\frac{\partial}{\partial x}(x) + \frac{\partial}{\partial x}\left(\sqrt{xy}\right) + \frac{\partial}{\partial x}(2\ln(z))$$

Apply the power rule $\frac{\partial}{\partial x}(x^n) = n\cdot x^{-1+n}$ with $n = 1$, in other words $\frac{\partial}{\partial x}(x) = 1$:

$$-\tan(z)\frac{\partial}{\partial x}(x) + \frac{\partial}{\partial x}\left(\sqrt{xy}\right) + \frac{\partial}{\partial x}(2\ln(z)) = -\tan(z)1 + \frac{\partial}{\partial x}\left(\sqrt{xy}\right) + \frac{\partial}{\partial x}(2\ln(z))$$

Write the function $\sqrt{xy}$ as a composition of the two functions $u = g = xy$ and $f(u) = \sqrt{u}$.

Apply the chain rule $\frac{\partial}{\partial x}(f(g)) = \frac{\partial}{\partial u}(f(u))\cdot\frac{\partial}{\partial x}(g)$:

$$-\tan(z) + \frac{\partial}{\partial x}\left(\sqrt{xy}\right) + \frac{\partial}{\partial x}(2\ln(z)) = -\tan(z) + \frac{\partial}{\partial u}(\sqrt{u})\,\frac{\partial}{\partial x}(xy) + \frac{\partial}{\partial x}(2\ln(z))$$

Apply the power rule $\frac{\partial}{\partial u}(u^n) = n\cdot u^{-1+n}$ with $n = \frac{1}{2}$:

$$-\tan(z) + \frac{\partial}{\partial u}(\sqrt{u})\frac{\partial}{\partial x}(xy) + \frac{\partial}{\partial x}(2\ln(z)) = -\tan(z) + \frac{\partial}{\partial u}\left(u^{\frac{1}{2}}\right)\frac{\partial}{\partial x}(xy) + \frac{\partial}{\partial x}(2\ln(z))$$
$$= -\tan(z) + \left(\frac{u^{-1+\frac{1}{2}}}{2}\right)\frac{\partial}{\partial x}(xy) + \frac{\partial}{\partial x}(2\ln(z)) = -\tan(z) + \left(\frac{u^{-\frac{1}{2}}}{2}\right)\frac{\partial}{\partial x}(xy)$$
$$+ \frac{\partial}{\partial x}(2\ln(z)) = -\tan(z) + \frac{\partial}{\partial x}(2\ln(z)) + \frac{\frac{\partial}{\partial x}(xy)}{2\sqrt{u}}$$

Return to the old variable:

$$-\tan(z)+\frac{\frac{1}{\sqrt{u}}\frac{\partial}{\partial x}(xy)}{2}+\frac{\partial}{\partial x}(2\ln(z))=-\tan(z)+\frac{\frac{1}{\sqrt{xy}}\frac{\partial}{\partial x}(xy)}{2}+\frac{\partial}{\partial x}(2\ln(z))$$

Apply the constant multiple rule $\frac{\partial}{\partial x}(c\cdot f)=c\cdot\frac{\partial}{\partial x}(f)$ with $c=y$ and $f=x$:

$$-\tan(z)+\frac{\partial}{\partial x}(2\ln(z))+\frac{\frac{\partial}{\partial x}(xy)}{2\sqrt{xy}}=-\tan(z)+\frac{\partial}{\partial x}(2\ln(z))+\frac{y\frac{\partial}{\partial x}(x)}{2\sqrt{xy}}$$

Apply the power rule $\frac{\partial}{\partial x}(x^n)=n\cdot x^{-1+n}$ with $n=1$, in other words $\frac{\partial}{\partial x}(x)=1$:

$$\frac{y\frac{\partial}{\partial x}(x)}{2\sqrt{xy}}-\tan(z)+\frac{\partial}{\partial x}(2\ln(z))=\frac{y1}{2\sqrt{xy}}-\tan(z)+\frac{\partial}{\partial x}(2\ln(z))$$

The derivative of a constant is 0:

$$\frac{y}{2\sqrt{xy}}-\tan(z)+\frac{\partial}{\partial x}(2\ln(z))=\frac{y}{2\sqrt{xy}}-\tan(z)+(0)$$

Thus, $\frac{\partial}{\partial x}\left(-x\tan(z)+\sqrt{xy}+3\ln(y)+2\ln(z)\right)=\frac{y}{2\sqrt{xy}}-\tan(z)$

NEXT, $\frac{\partial^2}{\partial x\partial y}\left(-x\tan(z)+\sqrt{xy}+3\ln(y)+2\ln(z)\right)$

$$=\frac{\partial}{\partial y}\left(\frac{\partial}{\partial x}\left(-x\tan(z)+\sqrt{xy}+3\ln(y)+2\ln(z)\right)\right)=\frac{\partial}{\partial y}\left(\frac{y}{2\sqrt{xy}}-\tan(z)\right)$$

The derivative of a sum/difference is the sum/difference of derivatives:

$$\frac{\partial}{\partial y}\left(\frac{y}{2\sqrt{xy}}-\tan(z)\right)=\left(\frac{\partial}{\partial y}\left(\frac{y}{2\sqrt{xy}}\right)-\frac{\partial}{\partial y}(\tan(z))\right)$$

Apply the constant multiple rule $\frac{\partial}{\partial y}(c \cdot f) = c \cdot \frac{\partial}{\partial y}(f)$ with $c = \frac{1}{2}$ and $f = \frac{y}{\sqrt{xy}}$:

$$\frac{\partial}{\partial y}\left(\frac{y}{2\sqrt{xy}}\right) - \frac{\partial}{\partial y}(\tan(z)) = \left(\frac{\frac{\partial}{\partial y}\left(\frac{y}{\sqrt{xy}}\right)}{2}\right) - \frac{\partial}{\partial y}(\tan(z))$$

Apply the quotient rule $\frac{\partial}{\partial y}\left(\frac{f}{g}\right) = \frac{1}{g^2}\left(\frac{\partial}{\partial y}(f) \cdot g - f \cdot \frac{\partial}{\partial y}(g)\right)$ with $f = y$ and $g = \sqrt{xy}$:

$$\frac{\frac{\partial}{\partial y}\left(\frac{y}{\sqrt{xy}}\right)}{2} - \frac{\partial}{\partial y}(\tan(z)) = \frac{\left(\frac{-y\frac{\partial}{\partial y}\left(\sqrt{xy}\right)+\frac{\partial}{\partial y}(y)\sqrt{xy}}{xy}\right)}{2} - \frac{\partial}{\partial y}(\tan(z))$$

Apply the power rule $\frac{\partial}{\partial y}(y^n) = n \cdot y^{-1+n}$ with $n = 1$, in other words $\frac{\partial}{\partial y}(y) = 1$:

$$-\frac{\partial}{\partial y}(\tan(z)) + \frac{-y\frac{\partial}{\partial y}\left(\sqrt{xy}\right) + \sqrt{xy}\frac{\partial}{\partial y}(y)}{2xy} = -\frac{\partial}{\partial y}(\tan(z)) + \frac{-y\frac{\partial}{\partial y}\left(\sqrt{xy}\right) + \sqrt{xy}1}{2xy}$$

The derivative of a constant is 0:

$$-\frac{\partial}{\partial y}(\tan(z)) + \frac{-y\frac{\partial}{\partial y}\left(\sqrt{xy}\right) + \sqrt{xy}}{2xy} = -(0) + \frac{-y\frac{\partial}{\partial y}\left(\sqrt{xy}\right) + \sqrt{xy}}{2xy}$$

Write the function $\sqrt{xy}$ as a composition of the two functions $u = g = xy$ and $f(u) = \sqrt{u}$.

Apply the chain rule $\frac{\partial}{\partial y}(f(g)) = \frac{\partial}{\partial u}(f(u)) \cdot \frac{\partial}{\partial y}(g)$:

$$\frac{-y\frac{\partial}{\partial y}\left(\sqrt{xy}\right) + \sqrt{xy}}{2xy} = \frac{-y\frac{\partial}{\partial u}\left(\sqrt{u}\right)\frac{\partial}{\partial y}(xy) + \sqrt{xy}}{2xy}$$

Apply the power rule $\frac{\partial}{\partial u}(u^n) = n \cdot u^{-1+n}$ with $n = \frac{1}{2}$:

$$\frac{-y \frac{\partial}{\partial u}(\sqrt{u}) \frac{\partial}{\partial y}(xy) + \sqrt{xy}}{2xy} = \frac{-y \frac{\partial}{\partial u}\left(u^{\frac{1}{2}}\right) \frac{\partial}{\partial y}(xy) + \sqrt{xy}}{2xy} = \frac{-y\left(\frac{u^{-1+\frac{1}{2}}}{2}\right) \frac{\partial}{\partial y}(xy) + \sqrt{xy}}{2xy}$$

$$= \frac{-y\left(\frac{u^{-\frac{1}{2}}}{2}\right) \frac{\partial}{\partial y}(xy) + \sqrt{xy}}{2xy} = \frac{\sqrt{xy} - \frac{y \frac{\partial}{\partial y}(xy)}{2\sqrt{u}}}{2xy}$$

Return to the old variable:

$$\frac{-\frac{y \frac{1}{\sqrt{u}} \frac{\partial}{\partial y}(xy)}{2} + \sqrt{xy}}{2xy} = \frac{-\frac{y \frac{1}{\sqrt{xy}} \frac{\partial}{\partial y}(xy)}{2} + \sqrt{xy}}{2xy}$$

Apply the constant multiple rule $\frac{\partial}{\partial y}(c \cdot f) = c \cdot \frac{\partial}{\partial y}(f)$ with $c = x$ and $f = y$:

$$\frac{-\frac{y \frac{\partial}{\partial y}(xy)}{2\sqrt{xy}} + \sqrt{xy}}{2xy} = \frac{-\frac{yx \frac{\partial}{\partial y}(y)}{2\sqrt{xy}} + \sqrt{xy}}{2xy}$$

Apply the power rule $\frac{\partial}{\partial y}(y^n) = n \cdot y^{-1+n}$ with $n = 1$, in other words $\frac{\partial}{\partial y}(y) = 1$:

$$\frac{-\frac{xy \frac{\partial}{\partial y}(y)}{2\sqrt{xy}} + \sqrt{xy}}{2xy} = \frac{-\frac{xy1}{2\sqrt{xy}} + \sqrt{xy}}{2xy} = \frac{1}{4\sqrt{xy}}$$

Thus, $\frac{\partial}{\partial y}\left(\frac{y}{2\sqrt{xy}} - \tan(z)\right) = \frac{1}{4\sqrt{xy}}$

Therefore, $\frac{\partial^2}{\partial x \partial y}\left(-x\tan(z) + \sqrt{xy} + 3\ln(y) + 2\ln(z)\right) = \frac{1}{4\sqrt{xy}}$

Answer: $\frac{\partial^2}{\partial x \partial y}\left(-x\tan(z) + \sqrt{xy} + \ln\left(y^3 z^2\right)\right) = \frac{1}{4\sqrt{xy}}$

See Here:
https://www.emathhelp.net/calculators/calculus-3/partial-derivative-calculator/?f=%28xy%29%5E0.5+%2B+ln+%28y%5E3+z%5E2%29+-x+tan+%28z%29&var=y%2C+x%2Cz

Your input: find $\frac{\partial^3}{\partial y \partial x \partial z}\left(-x\tan(z)+\sqrt{xy}+\ln\left(y^3z^2\right)\right)$

Rewrite the logarithm: $\frac{\partial^3}{\partial y \partial x \partial z}\left(-x\tan(z)+\sqrt{xy}+\ln\left(y^3z^2\right)\right)$

$$= \frac{\partial^3}{\partial y \partial x \partial z}\left(-x\tan(z)+\sqrt{xy}+3\ln(y)+2\ln(z)\right)$$

FIRST, FIND $\frac{\partial}{\partial y}\left(-x\tan(z)+\sqrt{xy}+3\ln(y)+2\ln(z)\right)$

The derivative of a sum/difference is the sum/difference of derivatives:

$$\frac{\partial}{\partial y}\left(-x\tan(z)+\sqrt{xy}+3\ln(y)+2\ln(z)\right)$$
$$=\left(\frac{\partial}{\partial y}\left(\sqrt{xy}\right)-\frac{\partial}{\partial y}(x\tan(z))+\frac{\partial}{\partial y}(3\ln(y))+\frac{\partial}{\partial y}(2\ln(z))\right)$$

Apply the constant multiple rule $\frac{\partial}{\partial y}(c\cdot f)=c\cdot\frac{\partial}{\partial y}(f)$ with $c=3$ and $f=\ln(y)$:

$$\frac{\partial}{\partial y}(3\ln(y))+\frac{\partial}{\partial y}\left(\sqrt{xy}\right)-\frac{\partial}{\partial y}(x\tan(z))+\frac{\partial}{\partial y}(2\ln(z))=\left(3\frac{\partial}{\partial y}(\ln(y))\right)+\frac{\partial}{\partial y}\left(\sqrt{xy}\right)$$
$$-\frac{\partial}{\partial y}(x\tan(z))+\frac{\partial}{\partial y}(2\ln(z))$$

The derivative of natural logarithm is $\frac{\partial}{\partial y}(\ln(y))=\frac{1}{y}$:

$$3\frac{\partial}{\partial y}(\ln(y))+\frac{\partial}{\partial y}\left(\sqrt{xy}\right)-\frac{\partial}{\partial y}(x\tan(z))+\frac{\partial}{\partial y}(2\ln(z))=3\frac{1}{y}+\frac{\partial}{\partial y}\left(\sqrt{xy}\right)$$
$$-\frac{\partial}{\partial y}(x\tan(z))+\frac{\partial}{\partial y}(2\ln(z))$$

The derivative of a constant is 0:

$$-\frac{\partial}{\partial y}(x\tan(z))+\frac{\partial}{\partial y}\left(\sqrt{xy}\right)+\frac{\partial}{\partial y}(2\ln(z))+\frac{3}{y}=-(0)+\frac{\partial}{\partial y}\left(\sqrt{xy}\right)+\frac{\partial}{\partial y}(2\ln(z))+\frac{3}{y}$$

Write the function $\sqrt{xy}$ as a composition of the two functions $u=g=xy$ and $f(u)=\sqrt{u}$.

Apply the chain rule $\frac{\partial}{\partial y}(f(g)) = \frac{\partial}{\partial u}(f(u)) \cdot \frac{\partial}{\partial y}(g)$:

$$\frac{\partial}{\partial y}(\sqrt{xy}) + \frac{\partial}{\partial y}(2\ln(z)) + \frac{3}{y} = \frac{\partial}{\partial u}(\sqrt{u})\,\frac{\partial}{\partial y}(xy) + \frac{\partial}{\partial y}(2\ln(z)) + \frac{3}{y}$$

Apply the power rule $\frac{\partial}{\partial u}(u^n) = n \cdot u^{-1+n}$ with $n = \frac{1}{2}$:

$$\frac{\partial}{\partial u}(\sqrt{u})\frac{\partial}{\partial y}(xy) + \frac{\partial}{\partial y}(2\ln(z)) + \frac{3}{y} = \frac{\partial}{\partial u}\left(u^{\frac{1}{2}}\right)\frac{\partial}{\partial y}(xy) + \frac{\partial}{\partial y}(2\ln(z)) + \frac{3}{y}$$
$$= \left(\frac{u^{-1+\frac{1}{2}}}{2}\right)\frac{\partial}{\partial y}(xy) + \frac{\partial}{\partial y}(2\ln(z)) + \frac{3}{y} = \left(\frac{u^{-\frac{1}{2}}}{2}\right)\frac{\partial}{\partial y}(xy) + \frac{\partial}{\partial y}(2\ln(z)) + \frac{3}{y}$$
$$= \frac{\partial}{\partial y}(2\ln(z)) + \frac{3}{y} + \frac{\frac{\partial}{\partial y}(xy)}{2\sqrt{u}}$$

Return to the old variable:

$$\frac{\frac{1}{\sqrt{u}}\frac{\partial}{\partial y}(xy)}{2} + \frac{\partial}{\partial y}(2\ln(z)) + \frac{3}{y} = \frac{\frac{1}{\sqrt{xy}}\frac{\partial}{\partial y}(xy)}{2} + \frac{\partial}{\partial y}(2\ln(z)) + \frac{3}{y}$$

Apply the constant multiple rule $\frac{\partial}{\partial y}(c \cdot f) = c \cdot \frac{\partial}{\partial y}(f)$ with $c = x$ and $f = y$:

$$\frac{\partial}{\partial y}(2\ln(z)) + \frac{\frac{\partial}{\partial y}(xy)}{2\sqrt{xy}} + \frac{3}{y} = \frac{\partial}{\partial y}(2\ln(z)) + \frac{x\frac{\partial}{\partial y}(y)}{2\sqrt{xy}} + \frac{3}{y}$$

Apply the power rule $\frac{\partial}{\partial y}(y^n) = n \cdot y^{-1+n}$ with $n = 1$, in other words $\frac{\partial}{\partial y}(y) = 1$:

$$\frac{x\frac{\partial}{\partial y}(y)}{2\sqrt{xy}} + \frac{\partial}{\partial y}(2\ln(z)) + \frac{3}{y} = \frac{x1}{2\sqrt{xy}} + \frac{\partial}{\partial y}(2\ln(z)) + \frac{3}{y}$$

The derivative of a constant is 0:

$$\frac{x}{2\sqrt{xy}} + \frac{\partial}{\partial y}\left(2\ln\left(z\right)\right) + \frac{3}{y} = \frac{x}{2\sqrt{xy}} + (0) + \frac{3}{y}$$

Thus, $\frac{\partial}{\partial y}\left(-x\tan\left(z\right) + \sqrt{xy} + 3\ln\left(y\right) + 2\ln\left(z\right)\right) = \frac{x}{2\sqrt{xy}} + \frac{3}{y}$

NEXT, $\frac{\partial^2}{\partial y \partial x}\left(-x\tan\left(z\right) + \sqrt{xy} + 3\ln\left(y\right) + 2\ln\left(z\right)\right)$

$$= \frac{\partial}{\partial x}\left(\frac{\partial}{\partial y}\left(-x\tan\left(z\right) + \sqrt{xy} + 3\ln\left(y\right) + 2\ln\left(z\right)\right)\right) = \frac{\partial}{\partial x}\left(\frac{x}{2\sqrt{xy}} + \frac{3}{y}\right)$$

The derivative of a sum/difference is the sum/difference of derivatives:

$$\frac{\partial}{\partial x}\left(\frac{x}{2\sqrt{xy}} + \frac{3}{y}\right) = \left(\frac{\partial}{\partial x}\left(\frac{3}{y}\right) + \frac{\partial}{\partial x}\left(\frac{x}{2\sqrt{xy}}\right)\right)$$

Apply the constant multiple rule $\frac{\partial}{\partial x}(c \cdot f) = c \cdot \frac{\partial}{\partial x}(f)$ with $c = \frac{1}{2}$ and $f = \frac{x}{\sqrt{xy}}$:

$$\frac{\partial}{\partial x}\left(\frac{x}{2\sqrt{xy}}\right) + \frac{\partial}{\partial x}\left(\frac{3}{y}\right) = \left(\frac{\frac{\partial}{\partial x}\left(\frac{x}{\sqrt{xy}}\right)}{2}\right) + \frac{\partial}{\partial x}\left(\frac{3}{y}\right)$$

Apply the quotient rule $\frac{\partial}{\partial x}\left(\frac{f}{g}\right) = \frac{1}{g^2}\left(\frac{\partial}{\partial x}(f) \cdot g - f \cdot \frac{\partial}{\partial x}(g)\right)$ with $f = x$ and $g = \sqrt{xy}$:

$$\frac{\frac{\partial}{\partial x}\left(\frac{x}{\sqrt{xy}}\right)}{2} + \frac{\partial}{\partial x}\left(\frac{3}{y}\right) = \frac{\left(\frac{-x\frac{\partial}{\partial x}\left(\sqrt{xy}\right) + \frac{\partial}{\partial x}(x)\sqrt{xy}}{xy}\right)}{2} + \frac{\partial}{\partial x}\left(\frac{3}{y}\right)$$

Apply the power rule $\frac{\partial}{\partial x}(x^n) = n \cdot x^{-1+n}$ with $n = 1$, in other words $\frac{\partial}{\partial x}(x) = 1$:

$$\frac{\partial}{\partial x}\left(\frac{3}{y}\right) + \frac{-x\frac{\partial}{\partial x}\left(\sqrt{xy}\right) + \sqrt{xy}\frac{\partial}{\partial x}(x)}{2xy} = \frac{\partial}{\partial x}\left(\frac{3}{y}\right) + \frac{-x\frac{\partial}{\partial x}\left(\sqrt{xy}\right) + \sqrt{xy}1}{2xy}$$

Write the function $\sqrt{xy}$ as a composition of the two functions $u = g = xy$ and $f(u) = \sqrt{u}$.

Apply the chain rule $\frac{\partial}{\partial x}(f(g)) = \frac{\partial}{\partial u}(f(u)) \cdot \frac{\partial}{\partial x}(g)$:

$$\frac{\partial}{\partial x}\left(\frac{3}{y}\right) + \frac{-x\frac{\partial}{\partial x}\left(\sqrt{xy}\right) + \sqrt{xy}}{2xy} = \frac{\partial}{\partial x}\left(\frac{3}{y}\right) + \frac{-x\frac{\partial}{\partial u}\left(\sqrt{u}\right)\frac{\partial}{\partial x}(xy) + \sqrt{xy}}{2xy}$$

Apply the power rule $\frac{\partial}{\partial u}(u^n) = n \cdot u^{-1+n}$ with $n = \frac{1}{2}$:

$$\frac{\partial}{\partial x}\left(\frac{3}{y}\right) + \frac{-x\frac{\partial}{\partial u}\left(\sqrt{u}\right)\frac{\partial}{\partial x}(xy) + \sqrt{xy}}{2xy} = \frac{\partial}{\partial x}\left(\frac{3}{y}\right) + \frac{-x\frac{\partial}{\partial u}\left(u^{\frac{1}{2}}\right)\frac{\partial}{\partial x}(xy) + \sqrt{xy}}{2xy}$$
$$= \frac{\partial}{\partial x}\left(\frac{3}{y}\right) + \frac{-x\left(\frac{u^{-1+\frac{1}{2}}}{2}\right)\frac{\partial}{\partial x}(xy) + \sqrt{xy}}{2xy} = \frac{\partial}{\partial x}\left(\frac{3}{y}\right) + \frac{-x\left(\frac{u^{-\frac{1}{2}}}{2}\right)\frac{\partial}{\partial x}(xy) + \sqrt{xy}}{2xy}$$
$$= \frac{\partial}{\partial x}\left(\frac{3}{y}\right) + \frac{\sqrt{xy} - \frac{x\frac{\partial}{\partial x}(xy)}{2\sqrt{u}}}{2xy}$$

Return to the old variable:

$$\frac{\partial}{\partial x}\left(\frac{3}{y}\right) + \frac{-\frac{x\frac{1}{\sqrt{u}}\frac{\partial}{\partial x}(xy)}{2} + \sqrt{xy}}{2xy} = \frac{\partial}{\partial x}\left(\frac{3}{y}\right) + \frac{-\frac{x\frac{1}{\sqrt{xy}}\frac{\partial}{\partial x}(xy)}{2} + \sqrt{xy}}{2xy}$$

Apply the constant multiple rule $\frac{\partial}{\partial x}(c \cdot f) = c \cdot \frac{\partial}{\partial x}(f)$ with $c = y$ and $f = x$:

$$\frac{\partial}{\partial x}\left(\frac{3}{y}\right) + \frac{-\frac{x\frac{\partial}{\partial x}(xy)}{2\sqrt{xy}} + \sqrt{xy}}{2xy} = \frac{\partial}{\partial x}\left(\frac{3}{y}\right) + \frac{-\frac{xy\frac{\partial}{\partial x}(x)}{2\sqrt{xy}} + \sqrt{xy}}{2xy}$$

Apply the power rule $\frac{\partial}{\partial x}(x^n) = n \cdot x^{-1+n}$ with $n = 1$, in other words $\frac{\partial}{\partial x}(x) = 1$:

$$\frac{\partial}{\partial x}\left(\frac{3}{y}\right) + \frac{-\frac{xy\frac{\partial}{\partial x}(x)}{2\sqrt{xy}} + \sqrt{xy}}{2xy} = \frac{\partial}{\partial x}\left(\frac{3}{y}\right) + \frac{-\frac{xy1}{2\sqrt{xy}} + \sqrt{xy}}{2xy}$$

The derivative of a constant is 0:

$$\frac{\partial}{\partial x}\left(\frac{3}{y}\right) + \frac{-\frac{xy}{2\sqrt{xy}} + \sqrt{xy}}{2xy} = (0) + \frac{-\frac{xy}{2\sqrt{xy}} + \sqrt{xy}}{2xy} = \frac{1}{4\sqrt{xy}}$$

Thus, $\frac{\partial}{\partial x}\left(\frac{x}{2\sqrt{xy}} + \frac{3}{y}\right) = \frac{1}{4\sqrt{xy}}$

NEXT, $\frac{\partial^3}{\partial y \partial x \partial z}\left(-x\tan(z) + \sqrt{xy} + 3\ln(y) + 2\ln(z)\right)$

$$= \frac{\partial}{\partial z}\left(\frac{\partial^2}{\partial y \partial x}\left(-x\tan(z) + \sqrt{xy} + 3\ln(y) + 2\ln(z)\right)\right) = \frac{\partial}{\partial z}\left(\frac{1}{4\sqrt{xy}}\right)$$

The derivative of a constant is 0:

$$\frac{\partial}{\partial z}\left(\frac{1}{4\sqrt{xy}}\right) = (0)$$

Thus, $\frac{\partial}{\partial z}\left(\frac{1}{4\sqrt{xy}}\right) = 0$

Therefore, $\frac{\partial^3}{\partial y \partial x \partial z}\left(-x\tan(z) + \sqrt{xy} + 3\ln(y) + 2\ln(z)\right) = 0$

Answer: $\frac{\partial^3}{\partial y \partial x \partial z}\left(-x\tan(z) + \sqrt{xy} + \ln\left(y^3 z^2\right)\right) = 0$

Spherical and Rectangular Coordinates

Convert spherical to cylindrical coordinates using a calculator.

Using Fig.1 below, the trigonometric ratios and Pythagorean theorem, it can be shown that the relationships between **spherical coordinates** (ρ, θ, ϕ) and **cylindrical coordinates** (r, θ, z) are as follows:

$r = \rho \sin\phi$, $\theta = \theta$, $z = \rho\cos\phi$ (I)

$\rho = \sqrt{r^2 + z^2}$, $\theta = \theta$, $\tan\phi = \dfrac{r}{z}$ (II)

with $0 \le \theta < 2\pi$ and $0 \le \phi \le \pi$

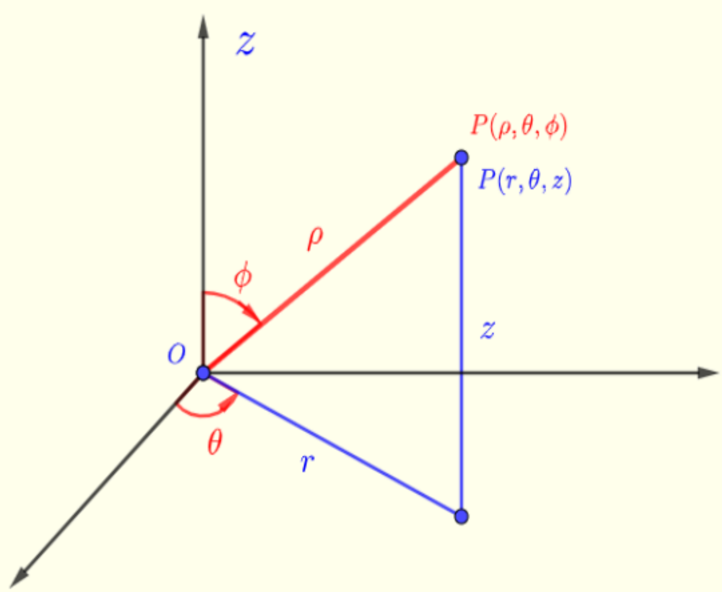

Fig.1 - Cylindrical and spherical coordinates

✅ Convert spherical to cylindrical:

Given spherical coordinates:

$$(\rho, \theta, \phi) = \left(5, \frac{4\pi}{3}, \frac{\pi}{6}\right)$$

Use the formulas:

$$r = \rho \sin\phi = 5 \cdot \sin\left(\frac{\pi}{6}\right) = 5 \cdot \frac{1}{2} = \boxed{2.5}$$

$$\theta = \frac{4\pi}{3}$$

$$z = \rho \cos\phi = 5 \cdot \cos\left(\frac{\pi}{6}\right) = 5 \cdot \frac{\sqrt{3}}{2} = \boxed{\frac{5\sqrt{3}}{2}}$$

So cylindrical coordinates:

$$(r, \theta, z) = \left(\boxed{2.5}, \boxed{\frac{4\pi}{3}}, \boxed{\frac{5\sqrt{3}}{2}}\right)$$

Rectangular and Spherical Coordinates

Convert rectangular to spherical coordinates using a calculator.

using simple trigonometry, it can be shown that the rectangular **rectangular coordinates** (x, y, z) and **spherical coordinates** (ρ, θ, ϕ) in Fig.1 are related as follows:

$x = \rho \sin\phi \cos\theta$, $y = \rho \sin\phi \sin\theta$, $z = \rho \cos\phi$ (I)

$\rho = \sqrt{x^2 + y^2 + z^2}$, $\tan\theta = \dfrac{y}{x}$, $\cos\phi = \dfrac{z}{\sqrt{x^2 + y^2 + z^2}}$ (II)

with $0 \le \theta < 2\pi$ and $0 \le \phi \le \pi$

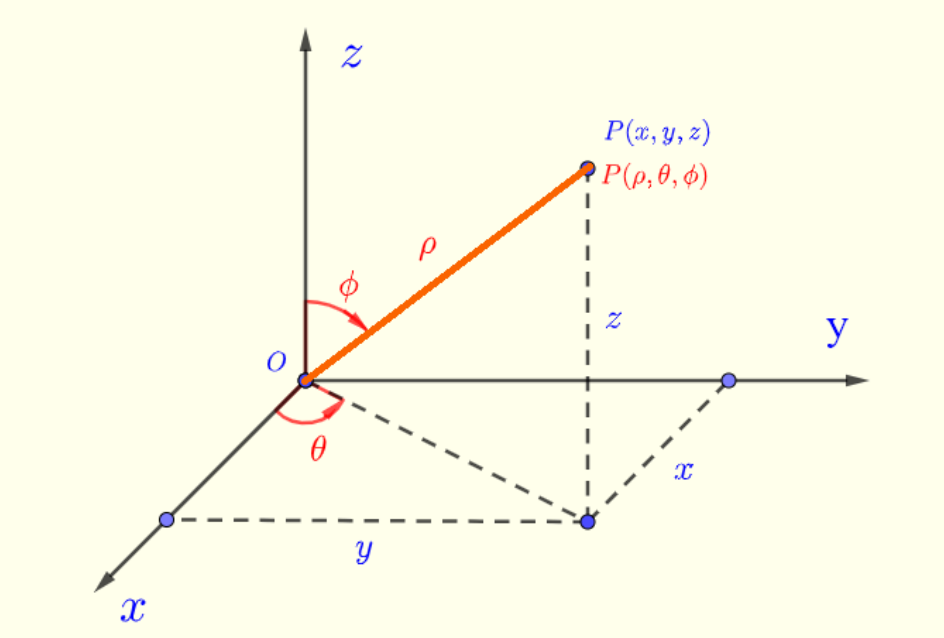

Fig.1 - Rectangular and spherical coordinates

✅ Convert rectangular to spherical:

Given Cartesian coordinates:

$$(x, y, z) = \left(4,\ -4,\ 4\sqrt{6}\right)$$

Use the formulas:

$$\rho = \sqrt{x^2 + y^2 + z^2} = \sqrt{16 + 16 + 96} = \sqrt{128} = \boxed{8\sqrt{2}}$$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{-4}{4}\right) = \tan^{-1}(-1) = \boxed{-\frac{\pi}{4}}$$

$$\phi = \cos^{-1}\left(\frac{z}{\rho}\right) = \cos^{-1}\left(\frac{4\sqrt{6}}{8\sqrt{2}}\right) = \cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = \boxed{\frac{\pi}{6}}$$

So spherical coordinates:

$$(\rho, \theta, \phi) = \left(\boxed{8\sqrt{2}}, \boxed{-\frac{\pi}{4}}, \boxed{\frac{\pi}{6}}\right)$$